

ALP Tabu Full Rerun Results

Run date: July 15, 2026

This report contains the complete rerun output for the Tabu recommendation study: four hard ALP cases, three seeds, and nine Tabu arms. The rerun result confirms the same best mean settings and best single runs as the first controlled study.

Source files

Rerun rows: tabu_method_push_package/results/tabu_recommendation_study_rerun_20260714/study_rows.csv
 Rerun summary: tabu_method_push_package/results/tabu_recommendation_study_rerun_20260714/study_summary.csv
 First study: ALP_Tabu_Handoff_2026-06-18/results/tabu_recommendation_study_20260624/study_rows.csv

Integrity check

Raw rerun rows: 108
 Summary rows: 36
 Cases: 4
 Arms: 9
 Seeds: 0, 1, 2
 Infeasible schedules: 0
 Row-level gap differences from first study: 2

Best mean setting by case

Case	Best arm	Mean	Median	Best	Runtime
airland11 m=2	fixed_t80	1.8651%	1.9047%	1.7146%	237.7s
airland12 m=2	fixed_t20	2.6549%	3.8540%	0.1121%	358.6s
airland13 m=2	fixed_t40	3.8922%	3.4400%	2.2032%	1278.8s
airland13 m=3	fixed_t40	4.9680%	5.3736%	3.4978%	1249.1s

Best single run by case

Case	Arm	Seed	Objective	BKS	Gap	Runtime
airland11 m=2	fixed_t80	0	1353.73	1330.91	1.7146%	237.7s
airland12 m=2	fixed_t20	1	1697.52	1695.62	0.1121%	366.4s
airland13 m=2	fixed_t40	1	4030.74	3943.85	2.2032%	1280.8s
airland13 m=3	fixed_t80	2	690.06	673.85	2.4056%	1249.6s

Complete 36-row summary table

Case	Arm	n	Mean	Median	Best	Std	Runtime
airland11 m=2	tabu_off	3	2.4209%	2.2834%	2.2834%	0.2382%	241.7s
airland11 m=2	fixed_t20	3	2.1426%	2.1684%	1.9761%	0.1553%	232.5s
airland11 m=2	fixed_t40	3	3.8375%	3.8297%	3.6005%	0.2409%	234.4s
airland11 m=2	fixed_t80	3	1.8651%	1.9047%	1.7146%	0.1352%	237.7s
airland11 m=2	fixed_t160	3	4.1485%	3.3090%	2.9904%	1.7373%	242.7s
airland11 m=2	iteration_t80	3	3.5782%	3.5855%	2.4006%	1.1740%	240.0s
airland11 m=2	fixed_t80_lp_aspiration	3	3.9006%	4.6900%	2.1391%	1.5282%	243.2s
airland11 m=2	fixed_t80_least_recent	3	1.9676%	1.9761%	1.9047%	0.0591%	249.4s
airland11 m=2	reactive_combined	3	2.9030%	2.6959%	2.2834%	0.7451%	239.3s
airland12 m=2	tabu_off	3	4.8449%	4.5559%	3.4082%	1.6008%	351.5s
airland12 m=2	fixed_t20	3	2.6549%	3.8540%	0.1121%	2.2033%	358.6s
airland12 m=2	fixed_t40	3	6.2115%	7.0482%	3.8723%	2.0530%	361.9s
airland12 m=2	fixed_t80	3	4.3789%	3.7178%	3.3410%	1.4834%	362.7s
airland12 m=2	fixed_t160	3	4.8079%	5.2984%	3.3410%	1.2934%	360.8s
airland12 m=2	iteration_t80	3	3.8193%	5.1598%	0.2784%	3.0965%	360.7s
airland12 m=2	fixed_t80_lp_aspiration	3	6.3511%	7.6591%	3.3410%	2.6142%	374.9s
airland12 m=2	fixed_t80_least_recent	3	4.3789%	3.7178%	3.3410%	1.4834%	384.6s
airland12 m=2	reactive_combined	3	5.7969%	5.4529%	5.2358%	0.7913%	369.0s
airland13 m=2	tabu_off	3	5.0267%	4.8904%	4.3303%	0.7737%	1281.9s
airland13 m=2	fixed_t20	3	4.7903%	4.5481%	4.4581%	0.4994%	1280.4s
airland13 m=2	fixed_t40	3	3.8922%	3.4400%	2.2032%	1.9547%	1278.8s
airland13 m=2	fixed_t80	3	5.3228%	4.9941%	3.8627%	1.6492%	1285.5s
airland13 m=2	fixed_t160	3	5.9298%	6.4587%	3.8627%	1.8600%	1286.3s
airland13 m=2	iteration_t80	3	5.6106%	5.4660%	4.5121%	1.1775%	1290.0s
airland13 m=2	fixed_t80_lp_aspiration	3	5.0734%	5.1954%	4.6176%	0.4087%	1287.4s
airland13 m=2	fixed_t80_least_recent	3	5.3228%	4.9941%	3.8627%	1.6492%	1289.1s
airland13 m=2	reactive_combined	3	5.1385%	4.5052%	4.1792%	1.3889%	1285.5s

airland13	m=3	tabu_off	3	6.3708%	6.0844%	5.4775%	1.0658%	1249.5s
airland13	m=3	fixed_t20	3	6.9348%	7.5506%	3.8109%	2.8660%	1245.3s
airland13	m=3	fixed_t40	3	4.9680%	5.3736%	3.4978%	1.3151%	1249.1s
airland13	m=3	fixed_t80	3	5.1040%	5.7401%	2.4056%	2.4433%	1246.6s
airland13	m=3	fixed_t160	3	6.7221%	7.2850%	5.3306%	1.2123%	1250.5s
airland13	m=3	iteration_t80	3	5.7174%	5.0456%	4.5559%	1.6064%	1254.6s
airland13	m=3	fixed_t80_lp_aspiration	3	5.4320%	5.3039%	4.1404%	1.3602%	1257.0s
airland13	m=3	fixed_t80_least_recent	3	5.1040%	5.7401%	2.4056%	2.4433%	1250.6s
airland13	m=3	reactive_combined	3	6.3080%	6.0444%	5.3291%	1.1340%	1250.2s

Rows that changed from the first saved study

Instance	m	Seed	Arm	Old gap	Rerun gap	Change
airland13	2	1	iteration_t80	2.9167%	5.4660%	+2.5493 pp
airland13	2	1	tabu_off	3.3706%	5.8595%	+2.4889 pp

Complete 108-row run table

Case		Seed	Arm	Objective	BKS	Gap	Feas	Runtime
airland11	m=2	0	tabu_off	1361.30	1330.91	2.2834%	True	233.7s
airland11	m=2	0	fixed_t20	1357.21	1330.91	1.9761%	True	238.3s
airland11	m=2	0	fixed_t40	1381.88	1330.91	3.8297%	True	242.8s
airland11	m=2	0	fixed_t80	1353.73	1330.91	1.7146%	True	237.7s
airland11	m=2	0	fixed_t160	1370.71	1330.91	2.9904%	True	241.1s
airland11	m=2	0	iteration_t80	1378.63	1330.91	3.5855%	True	236.1s
airland11	m=2	0	fixed_t80_lp_aspiration	1393.33	1330.91	4.6900%	True	239.7s
airland11	m=2	0	fixed_t80_least_recent	1357.82	1330.91	2.0219%	True	255.8s
airland11	m=2	0	reactive_combined	1361.30	1330.91	2.2834%	True	236.6s
airland11	m=2	1	tabu_off	1366.79	1330.91	2.6959%	True	260.4s
airland11	m=2	1	fixed_t20	1359.77	1330.91	2.1684%	True	226.4s
airland11	m=2	1	fixed_t40	1378.83	1330.91	3.6005%	True	227.2s
airland11	m=2	1	fixed_t80	1357.21	1330.91	1.9761%	True	225.2s
airland11	m=2	1	fixed_t160	1412.71	1330.91	6.1462%	True	225.7s
airland11	m=2	1	iteration_t80	1362.86	1330.91	2.4006%	True	225.2s
airland11	m=2	1	fixed_t80_lp_aspiration	1395.76	1330.91	4.8726%	True	228.5s
airland11	m=2	1	fixed_t80_least_recent	1357.21	1330.91	1.9761%	True	232.2s
airland11	m=2	1	reactive_combined	1380.55	1330.91	3.7298%	True	223.1s
airland11	m=2	2	tabu_off	1361.30	1330.91	2.2834%	True	231.1s
airland11	m=2	2	fixed_t20	1361.30	1330.91	2.2834%	True	232.8s
airland11	m=2	2	fixed_t40	1385.24	1330.91	4.0822%	True	233.3s
airland11	m=2	2	fixed_t80	1356.26	1330.91	1.9047%	True	250.3s
airland11	m=2	2	fixed_t160	1374.95	1330.91	3.3090%	True	261.2s
airland11	m=2	2	iteration_t80	1394.11	1330.91	4.7486%	True	258.8s
airland11	m=2	2	fixed_t80_lp_aspiration	1359.38	1330.91	2.1391%	True	261.3s
airland11	m=2	2	fixed_t80_least_recent	1356.26	1330.91	1.9047%	True	260.1s
airland11	m=2	2	reactive_combined	1366.79	1330.91	2.6959%	True	258.1s
airland12	m=2	0	tabu_off	1807.03	1695.62	6.5705%	True	362.6s
airland12	m=2	0	fixed_t20	1760.97	1695.62	3.8540%	True	364.9s
airland12	m=2	0	fixed_t40	1826.42	1695.62	7.7140%	True	369.9s
airland12	m=2	0	fixed_t80	1752.27	1695.62	3.3410%	True	368.9s
airland12	m=2	0	fixed_t160	1752.27	1695.62	3.3410%	True	369.1s
airland12	m=2	0	iteration_t80	1783.11	1695.62	5.1598%	True	372.8s
airland12	m=2	0	fixed_t80_lp_aspiration	1752.27	1695.62	3.3410%	True	371.7s
airland12	m=2	0	fixed_t80_least_recent	1752.27	1695.62	3.3410%	True	370.4s
airland12	m=2	0	reactive_combined	1788.08	1695.62	5.4529%	True	365.0s
airland12	m=2	1	tabu_off	1772.87	1695.62	4.5559%	True	354.6s
airland12	m=2	1	fixed_t20	1697.52	1695.62	0.1121%	True	366.4s
airland12	m=2	1	fixed_t40	1815.13	1695.62	7.0482%	True	371.1s
airland12	m=2	1	fixed_t80	1758.66	1695.62	3.7178%	True	371.9s
airland12	m=2	1	fixed_t160	1785.46	1695.62	5.2984%	True	365.6s
airland12	m=2	1	iteration_t80	1700.34	1695.62	0.2784%	True	365.7s
airland12	m=2	1	fixed_t80_lp_aspiration	1825.49	1695.62	7.6591%	True	406.5s
airland12	m=2	1	fixed_t80_least_recent	1758.66	1695.62	3.7178%	True	437.4s
airland12	m=2	1	reactive_combined	1784.40	1695.62	5.2358%	True	397.9s
airland12	m=2	2	tabu_off	1753.41	1695.62	3.4082%	True	337.3s
airland12	m=2	2	fixed_t20	1763.42	1695.62	3.9985%	True	344.6s

airland12	m=2	2	fixed_t40	1761.28	1695.62	3.8723%	True	344.8s
airland12	m=2	2	fixed_t80	1798.68	1695.62	6.0780%	True	347.3s
airland12	m=2	2	fixed_t160	1793.70	1695.62	5.7843%	True	347.8s
airland12	m=2	2	iteration_t80	1797.69	1695.62	6.0196%	True	343.6s
airland12	m=2	2	fixed_t80_lp_aspiration	1832.17	1695.62	8.0531%	True	346.6s
airland12	m=2	2	fixed_t80_least_recent	1798.68	1695.62	6.0780%	True	345.8s
airland12	m=2	2	reactive_combined	1809.26	1695.62	6.7020%	True	344.2s
airland13	m=2	0	tabu_off	4136.72	3943.85	4.8904%	True	1289.2s
airland13	m=2	0	fixed_t20	4123.22	3943.85	4.5481%	True	1277.4s
airland13	m=2	0	fixed_t40	4181.80	3943.85	6.0334%	True	1282.0s
airland13	m=2	0	fixed_t80	4096.19	3943.85	3.8627%	True	1288.8s
airland13	m=2	0	fixed_t160	4096.19	3943.85	3.8627%	True	1296.2s
airland13	m=2	0	iteration_t80	4121.80	3943.85	4.5121%	True	1291.4s
airland13	m=2	0	fixed_t80_lp_aspiration	4148.75	3943.85	5.1954%	True	1280.5s
airland13	m=2	0	fixed_t80_least_recent	4096.19	3943.85	3.8627%	True	1294.2s
airland13	m=2	0	reactive_combined	4209.32	3943.85	6.7312%	True	1278.8s
airland13	m=2	1	tabu_off	4174.94	3943.85	5.8595%	True	1276.5s
airland13	m=2	1	fixed_t20	4119.67	3943.85	4.4581%	True	1281.2s
airland13	m=2	1	fixed_t40	4030.74	3943.85	2.2032%	True	1280.8s
airland13	m=2	1	fixed_t80	4140.81	3943.85	4.9941%	True	1289.8s
airland13	m=2	1	fixed_t160	4198.57	3943.85	6.4587%	True	1274.5s
airland13	m=2	1	iteration_t80	4159.42	3943.85	5.4660%	True	1291.3s
airland13	m=2	1	fixed_t80_lp_aspiration	4125.96	3943.85	4.6176%	True	1282.8s
airland13	m=2	1	fixed_t80_least_recent	4140.81	3943.85	4.9941%	True	1295.5s
airland13	m=2	1	reactive_combined	4108.67	3943.85	4.1792%	True	1293.3s
airland13	m=2	2	tabu_off	4114.63	3943.85	4.3303%	True	1280.1s
airland13	m=2	2	fixed_t20	4155.42	3943.85	5.3646%	True	1282.7s
airland13	m=2	2	fixed_t40	4079.52	3943.85	3.4400%	True	1273.6s
airland13	m=2	2	fixed_t80	4224.32	3943.85	7.1116%	True	1277.8s
airland13	m=2	2	fixed_t160	4238.38	3943.85	7.4681%	True	1288.4s
airland13	m=2	2	iteration_t80	4214.15	3943.85	6.8537%	True	1287.4s
airland13	m=2	2	fixed_t80_lp_aspiration	4157.10	3943.85	5.4072%	True	1298.9s
airland13	m=2	2	fixed_t80_least_recent	4224.32	3943.85	7.1116%	True	1277.6s
airland13	m=2	2	reactive_combined	4121.53	3943.85	4.5052%	True	1284.5s
airland13	m=3	0	tabu_off	710.76	673.85	5.4775%	True	1251.8s
airland13	m=3	0	fixed_t20	699.53	673.85	3.8109%	True	1249.6s
airland13	m=3	0	fixed_t40	697.42	673.85	3.4978%	True	1253.7s
airland13	m=3	0	fixed_t80	712.53	673.85	5.7401%	True	1244.2s
airland13	m=3	0	fixed_t160	722.94	673.85	7.2850%	True	1242.0s
airland13	m=3	0	iteration_t80	707.85	673.85	5.0456%	True	1240.7s
airland13	m=3	0	fixed_t80_lp_aspiration	701.75	673.85	4.1404%	True	1258.8s
airland13	m=3	0	fixed_t80_least_recent	712.53	673.85	5.7401%	True	1245.7s
airland13	m=3	0	reactive_combined	714.58	673.85	6.0444%	True	1240.1s
airland13	m=3	1	tabu_off	714.85	673.85	6.0844%	True	1247.0s
airland13	m=3	1	fixed_t20	737.48	673.85	9.4428%	True	1239.2s
airland13	m=3	1	fixed_t40	714.50	673.85	6.0325%	True	1239.2s
airland13	m=3	1	fixed_t80	722.14	673.85	7.1663%	True	1245.8s
airland13	m=3	1	fixed_t160	709.77	673.85	5.3306%	True	1245.6s
airland13	m=3	1	iteration_t80	704.55	673.85	4.5559%	True	1262.1s
airland13	m=3	1	fixed_t80_lp_aspiration	720.02	673.85	6.8517%	True	1249.6s
airland13	m=3	1	fixed_t80_least_recent	722.14	673.85	7.1663%	True	1254.2s
airland13	m=3	1	reactive_combined	709.76	673.85	5.3291%	True	1252.6s
airland13	m=3	2	tabu_off	724.73	673.85	7.5506%	True	1249.5s
airland13	m=3	2	fixed_t20	724.73	673.85	7.5506%	True	1247.2s
airland13	m=3	2	fixed_t40	710.06	673.85	5.3736%	True	1254.3s
airland13	m=3	2	fixed_t80	690.06	673.85	2.4056%	True	1249.6s
airland13	m=3	2	fixed_t160	724.73	673.85	7.5506%	True	1264.0s
airland13	m=3	2	iteration_t80	724.73	673.85	7.5506%	True	1261.1s
airland13	m=3	2	fixed_t80_lp_aspiration	709.59	673.85	5.3039%	True	1262.6s
airland13	m=3	2	fixed_t80_least_recent	690.06	673.85	2.4056%	True	1251.8s
airland13	m=3	2	reactive_combined	724.73	673.85	7.5506%	True	1257.8s

Conclusion

The full rerun confirms the practical recommendation: keep the fixed-attribute Tabu memory and use

case-specific tenure. The supported baseline is fixed 80 for airland11 $m=2$, fixed 20 for airland12 $m=2$, and fixed 40 for airland13 by mean performance. For airland13 $m=3$, fixed 80 still gives the best single run. The adaptive variants did not win the mean gap on any case.